

```
mysql> /* set operators
/*> 1.union
/*> 2.union all
/*> 3.intersect
/*> 4.minus/except */
mysql> /* union:Combines results from two queries and removes duplicate rows.
Columns must have the same number and type.
/*> union all:Combines results from two queries and includes all rows, including
duplicates. Columns must have the same number and type.
/*> intersect: Returns rows that are common to both queries. Columns must have
the same number and type.
/*> Note: MySQL does not support INTERSECT directly.
/*> minus/except:Returns rows from the first query that are not present in the
second query. Columns must have the same number and type.
/*> */
```

```
mysql> drop table students;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE Students (
->     student_id INT PRIMARY KEY,
->     student_name VARCHAR(50),
->     course VARCHAR(50)
-> );
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
mysql> INSERT INTO Students VALUES
-> (101,'Rahul','Java Full Stack'),
-> (102,'Priya','Python Full Stack'),
-> (103,'Kiran','Data Science'),
-> (104,'Sneha','Java Full Stack'),
-> (105,'Ajay','Testing');
```

```
Query OK, 5 rows affected (0.01 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql>
mysql> CREATE TABLE Placed_Students (
->     student_id INT PRIMARY KEY,
->     student_name VARCHAR(50),
->     company_name VARCHAR(50)
-> );
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> INSERT INTO Placed_Students VALUES
-> (103,'Kiran','TCS'),
-> (104,'Sneha','Infosys'),
-> (105,'Ajay','Wipro'),
-> (106,'Pooja','Accenture'),
-> (107,'Ramesh','Capgemini');
```

Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from students;

student_id	student_name	course
101	Rahul	Java Full Stack
102	Priya	Python Full Stack
103	Kiran	Data Science
104	Sneha	Java Full Stack
105	Ajay	Testing

5 rows in set (0.00 sec)

mysql> select * from placed_students;

student_id	student_name	company_name
103	Kiran	TCS
104	Sneha	Infosys
105	Ajay	Wipro
106	Pooja	Accenture
107	Ramesh	Capgemini

5 rows in set (0.00 sec)

mysql> select student_name from students union select student_name from placed_students;

student_name
Rahul
Priya
Kiran
Sneha
Ajay
Pooja
Ramesh

7 rows in set (0.01 sec)

mysql> -- union all

mysql> select student_name from students union all select student_name from placed_students;

student_name
Rahul
Priya

Kiran
Sneha
Ajay
Kiran
Sneha
Ajay
Pooja
Ramesh

+-----+

10 rows in set (0.00 sec)

mysql> -- INTERSECT

mysql> select student_name from students intersect select student_name from placed_students;

+-----+

student_name
Kiran
Sneha
Ajay

+-----+

3 rows in set (0.00 sec)

mysql> -- minus/except

mysql> select student_name from students except select student_name from placed_students;

+-----+

student_name
Rahul
Priya

+-----+

2 rows in set (0.00 sec)

mysql> /* joins

/*> a join in MySQL is used to combine rows from two or more tables based on related columns.

/*> example

/*> employees table--->Employee details.

/*> departments table---> department details.

/*>

/*> if you want to see which employee belongings to which department. you need to join these two tables.

/*>

/*> Why do we need joins?

/*> -----

/*> 1. to fetch data from multiple tables in a single Query.

/*> 2. to reduce data duplication(instead of keeping everything in one table.)

/*> 3. to implement relationships like(one-to-one, one-to-many, many-to-one,

many-to-many) in databases.

```
/*> 4. to generate reports and analytics combining different datasets.
```

```
/*>
```

```
/*> types of joins in MySQL
```

```
/*> -----
```

```
/*> 1.inner join
```

```
/*> 2.left join
```

```
/*> 3. right join
```

```
/*> 4. full join
```

```
/*> 5. cross join
```

```
/*> 6. self join
```

```
/*> */
```

```
mysql> -- 1. INNER JOIN:Returns only the rows that have matching values in both tables.
```

```
mysql> -- 2. LEFT JOIN:Returns all rows from the left table and the matched rows from the right table. If no match, NULL values appear for right table columns.
```

```
mysql> -- 3.RIGHT JOIN:Returns all rows from the right table and the matched rows from the left table. If no match, NULL values appear for left table columns
```

```
mysql> -- 4. FULL JOIN:To simulate, use the UNION of LEFT JOIN and RIGHT JOIN.
```

```
mysql> -- 5.CROSS JOIN:Returns Cartesian product ? every row from the first table with every row from the second table.
```

```
mysql> -- 6. SELF JOIN:A table joined with itself.
```

```
mysql>
```

```
mysql> create table tab1(numid int);
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> create table tab2(numid int);
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> insert into tab1 values(10),(11),(12),(14);
```

```
Query OK, 4 rows affected (0.01 sec)
```

```
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> insert into tab2 values(11),(12),(13),(15);
```

```
Query OK, 4 rows affected (0.01 sec)
```

```
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> select * from tab1;
```

```
+-----+
```

```
| numid |
```

```
+-----+
```

```
|    10 |
```

```
|    11 |
```

```
|    12 |
```

```
|    14 |
```

```
+-----+
```

```
4 rows in set (0.00 sec)
```

```
mysql> select * from tab2;
```

```
+-----+
```

numid
11
12
13
15

4 rows in set (0.00 sec)

```
mysql> -- inner join
```

```
mysql> select * from tab1 inner join tab2 on tab1.numid = tab2.numid;
```

numid	numid
11	11
12	12

2 rows in set (0.00 sec)

```
mysql> select * from tab1 left join tab2 on tab1.numid = tab2.numid;
```

numid	numid
10	NULL
11	11
12	12
14	NULL

4 rows in set (0.00 sec)

```
mysql> select * from tab1 right join tab2 on tab1.numid = tab2.numid;
```

numid	numid
11	11
12	12
NULL	13
NULL	15

4 rows in set (0.00 sec)

```
mysql> select * from tab1 left join tab2 on tab1.numid = tab2.numid;
```

numid	numid
10	NULL
11	11
12	12
14	NULL

4 rows in set (0.00 sec)

```
mysql> select * from tab1 left join tab2 on tab1.numid = tab2.numid
-> union
-> select * from tab1 right join tab2 on tab1.numid = tab2.numid;
```

numid	numid
10	NULL
11	11
12	12
14	NULL
NULL	13
NULL	15

6 rows in set (0.00 sec)

```
mysql> select * from tab1 cross join tab2;
```

numid	numid
14	11
12	11
11	11
10	11
14	12
12	12
11	12
10	12
14	13
12	13
11	13
10	13
14	15
12	15
11	15
10	15

16 rows in set (0.00 sec)

```
mysql> notee
```